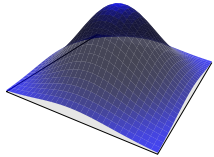
TECHNISCHE UNIVERSITÄT WIEN



Lecture 4

Statistics and Probability Theory

Continuous Probability Spaces and Random Variables

Daniel Kapla[1] and Andrey Kofnov[2]

E105 – Institute of Statistics and Mathematical Methods in Economics

[1]daniel.kapla@tuwien.ac.at, [2]andrey.kofnov@tuwien.ac.at

Continuous Probability Space

$\Omega \subseteq \mathbb{R}$

A *continuous* probability space $(\Omega, \mathfrak{S}, P)$ is given if Ω *is not countable*.

The classic setting is $\Omega \subseteq \mathbb{R}$ being a (union of) interval(s). For example;

$$\Omega = \mathbb{R}, \Omega = [0, 1], \Omega = [0, \infty), \Omega = (-1, 1), \Omega = (-\infty, -1] \cup [1, \infty), ...$$

$\mathfrak{S}$ conains all open intervals in Ω, the smallest is called Borel σ-algebra

Elementary events can not be given in a table! A different approach is needed. We will not go into details, we take a more practical approach via random variables.

Random Variables

A *Random Variable* is a *real valued function of elementary events*.[2]

$$X : \Omega \to \mathbb{R}$$

where every elementary event $\omega \in \Omega$ gets a real number $X(\omega) = x \in \mathbb{R}$ assigned.

- The number $x \in \mathbb{R}$ is one *realization of* X.
- The set $\{X(\omega) : \omega \in \Omega\}$ of *all values of* X is the *feature space* or *image of* X.
- *Events* are given by *pre-images* $X^{-1}(\mathcal{X}) = \{\omega \in \Omega : X(\Omega) \in \mathcal{X}\}$ for $\mathcal{X} \subseteq \mathbb{R}$. Thats how probabilities are assigned! But, the notation is cumbersome.
- We use a more describtive notation, for example; $[X = x] = X^{-1}(\{x\})$, $[X < 0] = X^{-1}((-\infty, 0))$, $[|X| \leq 1] = X^{-1}([-1, 1])$
- In practice, random variables are the main tool.

[2] More presicely, it is a measurable function from a probability space into a measure space.

Random Variables

There are *two types of random variables*[3]

Discrete

If the *image of* X is *finite or countable* the random variable X is discrete.

Example:
Number of children born next Tuesday

Continuous

If the *image of* X is *un-countable* (e.g. real intervals) the random variable X is continuous.

Example:
Time of the first child born next Tuesday

[3] ignoring mixtures

Discrete Random Variables

The *probability mass function* (pmf) f of a discrete random variable X is

$$f(x) = P(X = x) = P(X^{-1}(\{x\}))$$

PROPERTIES:

- It holds $0 \leq f(x) \leq 1$ for all $x \in \mathbb{R}$
- If X never takes the values x, then $f(x) = 0$

For example, let *X be the maximum of two 6-sided dice.*
We describe this by listing all values and the associated probabilities

Probability Table

values	x	1	2	3	4	5	6
pmf	$f(x)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

$$\sum_{x=1}^{6} f(x) = 1$$

Continuous Random Variables

A *random variable X is continuous* if a function $f : \mathbb{R} \to [0, \infty)$ *exists*, such that

$$P(a \leq X \leq b) = \int_a^b f(x)\,\mathrm{d}x, \qquad \text{for all } a, b \in \mathbb{R} : a \leq b.$$

The function f *is the probability density function* (*p.d.f.* or *density*.)

PROPERTIES:

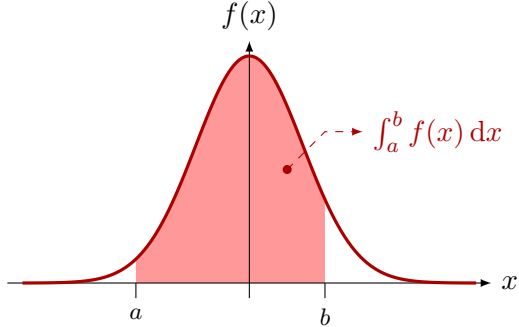
- The density f is *non-negative*, that is $f(x) \geq 0$ for all $x \in \mathbb{R}$
  It is not a probability, the density can take values bigger than 1.
- If X never takes the value x, then $f(x) = 0$
- The *density integrates to* 1

$$\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$$

Continuous Random Variables

For a *continuous random variable X with density f*, the *probability of* an event $[a \leq X \leq b]$ is the *area under the density*

$$P(a \leq X \leq b) = \int_a^b f(x)\,\mathrm{d}x$$



Since for any $a \in \mathbb{R}$

$$P(X = a) = P(a \leq X \leq a) = \int_a^a f(x)\,\mathrm{d}x = 0$$

it follows that for any $a, b \in \mathbb{R} : a \leq b$

$$P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b) = \int_a^b f(x)\,\mathrm{d}x.$$

Cumulative Distribution Function

Uniquely identifies Random Variables

The *cumulative distribution function* F (cdf) of a random variable (continuous or discrete) X is defined as

$$F(x) = P(X \leq x), \qquad \text{for all } x \in \mathbb{R}.$$

PROPERTIES:

- The cdf F is *bounded*; $0 \leq F(x) \leq 1$ for all $x \in \mathbb{R}$
- F is *monotonically increasing*, that is $x \leq y$ implies $F(x) \leq F(y)$
- It is *right-continuous*

$$\lim_{h \to 0+} F(x+h) = F(x), \qquad \text{for all } x \in \mathbb{R}$$

- and the left/right asymtotics are

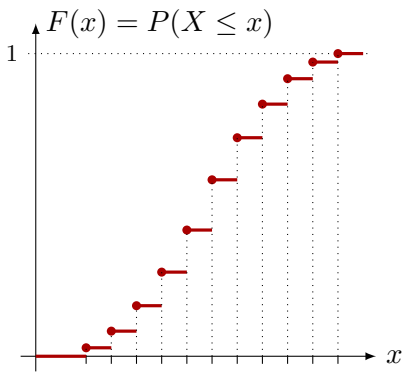
$$\lim_{x \to -\infty} F(x) = 0, \qquad \lim_{x \to +\infty} F(x) = 1.$$

Cumulative Distribution Function

Uniquely identifies Random Variables

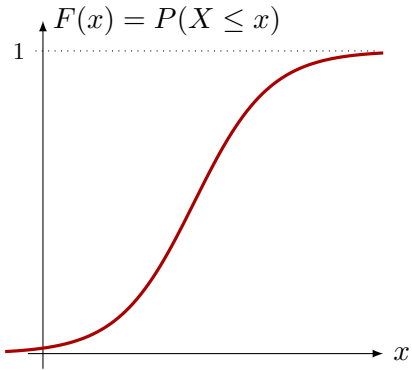
Discrete

The cdf F is a *step-function*



Continuous

The cdf F is *continuous*



Cumulative Distribution Function

Uniquely identifies Random Variables

For *discrete* random variables the *pmf* takes the same role as the *density* for *continuous* random variables.

Discrete

Let X be a discrete r.v. taking values $x_1, x_2, \ldots$ with pmf f and cdf F, then

$$F(x) = P(X \leq x) = \sum_{\substack{i=1 \\ x_i \leq x}}^{\infty} f(x_i)$$

We get $P(a < X \leq b) = F(b) - F(a)$

Continuous

Let X be a continuous random variable with density f and cdf F, then

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(z)\,\mathrm{d}z$$

We get $P(a < X \leq b) = F(b) - F(a)$, and the *density is the cdf's derivative*

$$f(x) = F'(x)$$

Expected Value – Definition

The *expected value* is a *weighted mean* (average) of the possible outcomes of X

Expectation for Discrete Random Variables

The *expected value* $\mathbb{E}[X]$ of a *discrete random variable* X with values $x_1, x_2, \ldots$ and *probability mass function* f is defined as

$$\mathbb{E}[X] = \sum_{i=1}^{\infty} x_i f(x_i) = x_1 \cdot f(x_1) + x_2 \cdot f(x_2) + \ldots$$

Expectation for Continuous Random Variables

The *expected value* $\mathbb{E}[X]$ of a *continuous random variable* X with *density* f is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) \, \mathrm{d}x$$

It is the *first* (raw) *moment of* X.

Absolute convergence assumed, else the expectation may be $\pm\infty$ or does not exist!

Expected Value – Properties

The *expectation is linear*, that is for two random variables X, Y and $a, b \in \mathbb{R}$ holds

$$\mathbb{E}[aX + b] = a\,\mathbb{E}\,X + b$$
$$\mathbb{E}[aX + bY] = a\,\mathbb{E}\,X + b\,\mathbb{E}\,Y$$

If X and Y are *independent*, then

$$\mathbb{E}[X \cdot Y] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$$

For a measurable function g (assuming absolute convergence) holds

$$\mathbb{E}[g(X)] = \sum_{i=1}^{\infty} g(x_i) f(x_i) \qquad \text{or} \qquad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f(x)\,\mathrm{d}x$$

and the most common/named cases are

k-TH (RAW) MOMENT: $E[X^k]$

k-TH CENTRAL MOMENT: $\mathbb{E}[(X - \mathbb{E}\,X)^k]$

VARIANCE: $\mathbb{V}\mathrm{ar}(X) = \mathbb{E}[(X - \mathbb{E}\,X)^2]$

STANDARD DEVIATION: $\sigma = \sqrt{\mathbb{V}\mathrm{ar}(X)}$

Variance – Properties

The *Variance* is a *measure of dispersion*, with random variables X, Y and $a, b \in \mathbb{R}$

$$\begin{aligned} \mathbb{V}\mathrm{ar}(X) &= \mathbb{E}[(X - \mathbb{E}\,X)^2] = \mathbb{E}[X^2] - (\mathbb{E}\,X)^2, \\ \mathbb{V}\mathrm{ar}(aX + b) &= a^2\,\mathbb{V}\mathrm{ar}(X). \end{aligned}$$

If X and Y are *independent*, then $(\mathbb{C}\mathrm{ov}(X, Y) = 0)$

$$\mathbb{V}\mathrm{ar}(X + Y) = \mathbb{V}\mathrm{ar}(X) + \mathbb{V}\mathrm{ar}(Y)$$

but if X and Y are not independent, then

$$\mathbb{V}\mathrm{ar}(X + Y) = \mathbb{V}\mathrm{ar}(X) + \mathbb{V}\mathrm{ar}(Y) + 2\,\mathbb{C}\mathrm{ov}(X, Y)$$

where $\mathbb{C}\mathrm{ov}(X, Y)$ is the *covariance* defind as

$$\mathbb{C}\mathrm{ov}(X, Y) = \mathbb{E}[(X - \mathbb{E}\,X)(Y - \mathbb{E}\,Y)]$$

which is a *measure of alignement and joint variability*.

Generalized Inverse and Quantile Functions

One often needs the *inverse* function of a cdf F.
The cdf is monotonically increasing but *not guaranteed to be one-to-one*!

Therefore, we need the *generalized inverse* of F

$$F^{-1}(p) = \inf\{x : F(x) \geq p\} \qquad \text{for } p \in (0, 1).$$

if F is invertible, then F^{-1} is the "usual" inverse of F.

The generalized inverse F^{-1} *of the cdf* F is known as the *quantile function*.

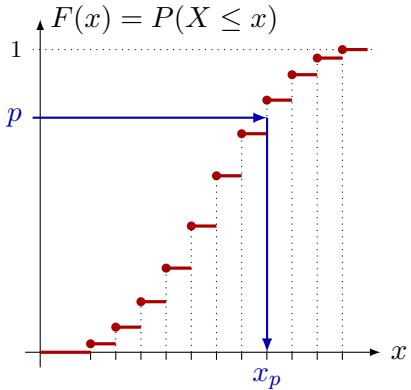
$$x_p = F^{-1}(p) \qquad \Leftrightarrow \qquad F(x_p) = p$$

for all $p \in (0, 1)$. The value x_p is a *p-quantile* (quantiles are in general not unique).

Generalized Inverse and Quantile Functions

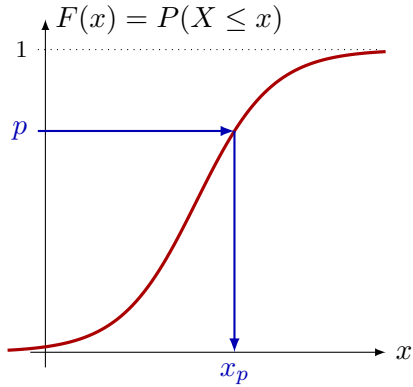
Discrete

p-quantile $x_p = F^{-1}(p)$ for $p \in (0, 1)$



Continuous

p-quantile $x_p = F^{-1}(p)$ for $p \in (0, 1)$



Splitting by Probability

Some special *quantiles* are the *quartiles*.

LOWER QUARTILE: is a 25%-quantile

MEDIAN: is a 50%-quantile

The *median splits a random variables image* into *two equally likely parts*.

UPPER QUARTILE: is a 75%-quantile

Another variation are the *percentiles*, which are in steps of $\frac{1}{100}$. For example a 0.1-quantile is the same as a 10-percentile. remember $0.1 = 10\%$

Transformation Theorem

Given a *continuous random variable* X with *density* f_X and a *function* $g : \mathbb{R} \to \mathbb{R}$
What is the *density* f_Y *of the transformed random variable* $Y = g(X)$?

METHOD 1: First compute the cdf F_Y for $Y = g(X)$ and then $f_Y = F_Y'$.

METHOD 2: Use the

DENSITY TRANSFORMATION THEOREM

If g is a *one-to-one* function with *differentiable inverse* g^{-1}, then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{\mathrm{d}}{\mathrm{d}y} g^{-1}(y) \right|.$$

Bernoulli distribution

The *Bernoulli distribution* models any setting with *two outcomes*, commonly denoted as *success vs. failure* parameterized with *success probability* $p \in [0, 1]$.

$$f(x) = p^x (1 - p)^{1-x}$$

for $x \in \{0, 1\}$ with $x = 1$ *a success* is its pmf. We write $X \sim \text{Bernoulli}(p)$.

The *expectation* of X is

$$\mathbb{E}[X] = \sum_{x=0}^{1} x f(x) = 0 \cdot f(0) + 1 \cdot f(1) = 0 + p^1 (1 - p)^0 = p.$$

Bernoulli distribution

The *Bernoulli distribution* models any setting with *two outcomes*, commonly denoted as *success vs. failure* parameterized with *success probability* $p \in [0, 1]$.

$$f(x) = p^x (1-p)^{1-x}$$

for $x \in \{0, 1\}$ with $x = 1$ *a success* is its pmf. We write $X \sim \text{Bernoulli}(p)$.

The *variance* of X, using $\mathbb{E}[X] = p$ is

$$\begin{aligned}\mathbb{V}\text{ar}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \sum_{x=0}^{1} (x - \mathbb{E}[X])^2 f(x) &= p^2 f(0) + (1-p)^2 f(1) \\ &= p^2(1-p) + (1-p)^2 p = p(1-p)\end{aligned}$$

Bernoulli distribution

The *Bernoulli distribution* models any setting with *two outcomes*, commonly denoted as *success vs. failure* parameterized with *success probability* $p \in [0, 1]$.

$$f(x) = p^x (1-p)^{1-x}$$

for $x \in \{0, 1\}$ with $x = 1$ *a success* is its pmf. We write $X \sim \text{Bernoulli}(p)$.

The cdf $F_X : \mathbb{R} \to [0, 1]$ is then

$$F_X(x) = P(X \leq x) = \begin{cases} 0, & x < 0 \\ f(0) = 1 - p, & 0 \leq x < 1 \\ f(0) + f(1) = 1, & 1 \leq x \end{cases}$$

Bernoulli distribution

The *Bernoulli distribution* models any setting with *two outcomes*, commonly denoted as *success vs. failure* parameterized with *success probability* $p \in [0, 1]$.

$$f(x) = p^x (1-p)^{1-x}$$

for $x \in \{0, 1\}$ with $x = 1$ *a success* is its pmf. We write $X \sim \text{Bernoulli}(p)$.

Lets now consider $y = g(x) = 2x - 1 \in \{-1, +1\}$ for $x \in \{0, 1\}$.
What is the cdf F_Y of $Y = g(X)$?

$$F_Y(y) = P(Y \leq y) = P(2X - 1 \leq y) = P(X \leq (y+1)/2))$$

$$= F_X((y+1)/2)) = \begin{cases} 0, & y < -1 \\ 1-p, & -1 \leq y < 1 \\ 1, & 1 \leq y \end{cases}$$

Bernoulli distribution

The *Bernoulli distribution* models any setting with *two outcomes*, commonly denoted as *success vs. failure* parameterized with *success probability* $p \in [0, 1]$.

$$f(x) = p^x (1-p)^{1-x}$$

for $x \in \{0, 1\}$ with $x = 1$ *a success* is its pmf. We write $X \sim \text{Bernoulli}(p)$.

Lets now consider $y = g(x) = 2x - 1 \in \{-1, +1\}$ for $x \in \{0, 1\}$.
What is the pdf f_Y of $Y = g(X)$?

Similarly, for $y \in \{-1, +1\}$ we get

$$\begin{aligned} f_Y(y) = P(Y = y) = P(2X - 1 = y) = P(X = (y+1)/2)) \\ = f_X((y+1)/2)) = p^{(y+1)/2}(1-p)^{1-(y+1)/2} \end{aligned}$$

Binomial distribution

The *Binomial distribution* $\mathrm{Bin}(n, p)$ models the *number of successes* in n *independent* Bernoulli trials with *success probability* p. The pmf is

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

for $x \in \{0, 1, \ldots, n\}$ with $p \in [0, 1]$.

EXPECTATION: We can write X *as the sum of* $B_i \overset{\text{iid}}{\sim} \mathrm{Bernoulli}(p)$ and use the *linearity of the expectation*

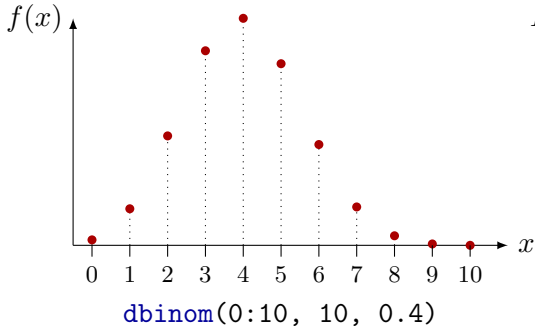
$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}\, B_i = \sum_{i=1}^{n} p = np$$

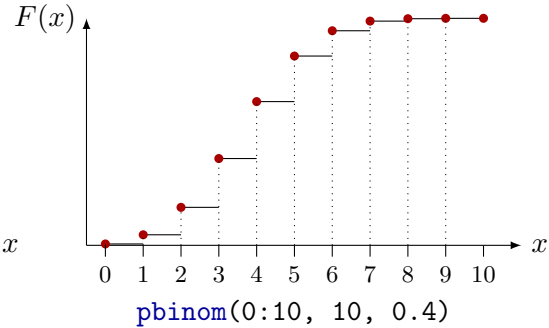
VARIANCE: Using that X is a sum of *independent* and identically distributed *Bernoulli* gives

$$\mathbb{V}\mathrm{ar}(X) = \sum_{i=1}^{n} \mathbb{V}\mathrm{ar}(B_i) = np(1-p)$$

Binomial distribution

For example the pmf f and cdf F of $X \sim \text{Bin}(10, 0.4)$;





[3] Use `qbinom` for the quantile function and `rbinom` to sample from the Binomial.

Geometric distribution

The *Geometric distribution* models the *number of attempts before* (or untill) *a success with probability p*. The pmf is

$$f(x) = (1-p)^x p$$

for $x \in \{0, 1, 2, \ldots\}$.

Expectation

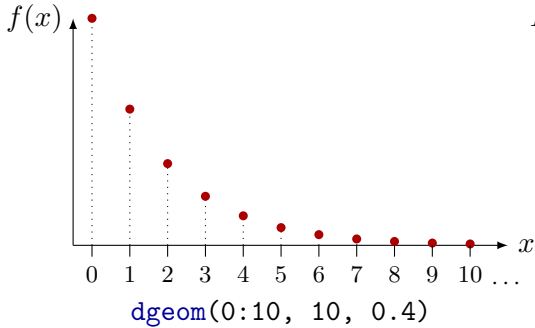
$$\mathbb{E}[X] = \frac{1-p}{p}$$

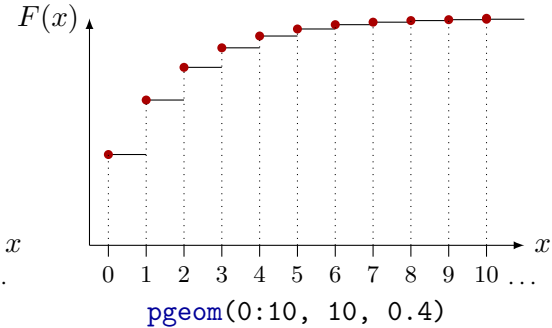
Variance

$$\mathbb{V}\text{ar}(X) = \frac{1-p}{p^2}$$

Geometric distribution

For example the pmf f and cdf F of $X \sim \text{Geom}(0.4)$;





[3] Use `qgeom` for the quantile function and `rgeom` to sample from the Geometric distribution.

Poisson distribution

The *Poisson distribution* models the *number of occurences in a time span* with pmf

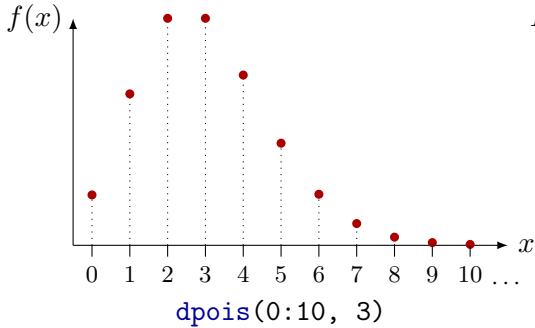
$$f(x) = e^{-\lambda}\frac{\lambda^x}{x!}$$

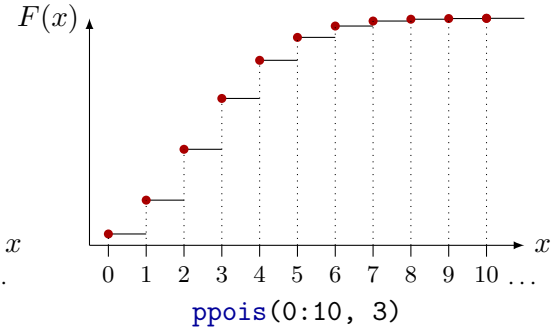
where $\lambda \in (0, \infty)$.

Expectation	Variance
$\mathbb{E}[X] = \lambda$	$\mathbb{V}\text{ar}(X) = \lambda$

Poisson distribution

For example the pmf f and cdf F of $X \sim \text{Poisson}(3)$;





[3] Use `qpois` for the quantile function and `rpois` to sample from the Poisson.

Uniform Distribution

The *density f of the Uniform distribution* for $X \sim \mathrm{U}(a, b)$ for $a, b \in \mathbb{R}$, $a < b$ is

$$f(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a < x < b \\ 0, & \text{else} \end{cases}$$

EXPECTATION: By integration

$$\begin{aligned} \mathbb{E}[X] &= \int_a^b x f(x)\,\mathrm{d}x \\ &= \int_a^b \frac{x}{b-a}\,\mathrm{d}x = \frac{a+b}{2} \end{aligned}$$

VARIANCE: Also by integration

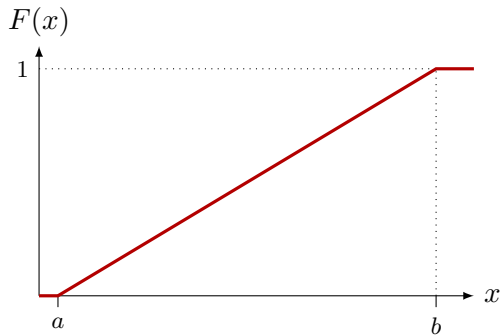
$$\begin{aligned} \mathbb{V}\mathrm{ar}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \int_a^b (x - \mathbb{E}[X])^2 f(x)\,\mathrm{d}x \\ &= \frac{1}{4(b-a)} \int_a^b (2x - a - b)^2\,\mathrm{d}x = \frac{(b-a)^2}{12} \end{aligned}$$

Uniform distribution

The density f and cdf F of $X \sim \mathrm{U}(a, b)$ for $a < b$;



`dunif(x, a, b)`



`punif(x, a, b)`

[3] Use `qunif` for the quantile function and `runif` to sample from the Uniform.

Exponential Distribution

The *Exponential distribution* models *waiting times*[4] with *density* f given as

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{for } x \geq 0, \\ 0, & \text{else.} \end{cases}$$

for $\lambda \in (0, \infty)$.

Expectation

$$\mathbb{E}[X] = \frac{1}{\lambda}$$

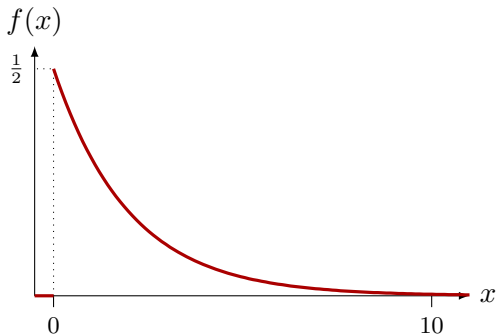
Variance

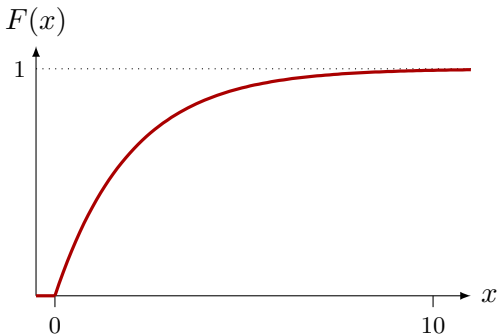
$$\mathbb{V}\text{ar}(X) = \frac{1}{\lambda^2}$$

[4] continuous analogue to the Geometric distribution

Exponential distribution

For example the density f and cdf F of $X \sim \text{Exp}(1/2)$;





[4] Use `qexp` for the quantile function and `rexp` to sample from the Exponential.

Normal distribution

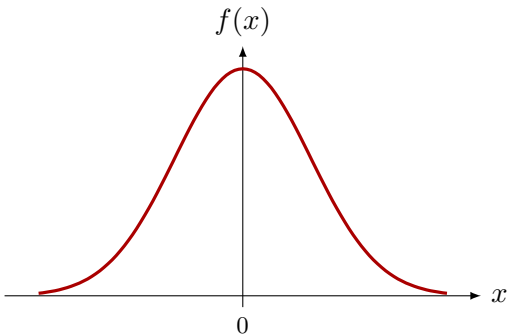
One of the most important distributions is the *normal (or Gaussian) distribution*

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big)$$

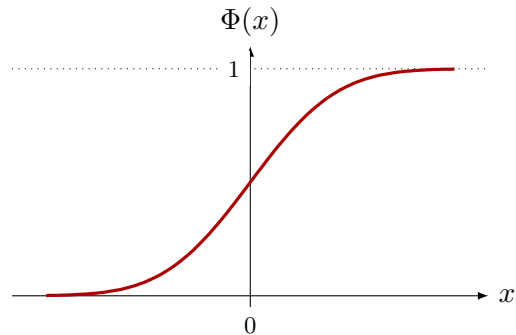
parmeterized by its mean $\mu \in \mathbb{R}$ and its variance $\sigma^2 > 0$.

Normal distribution

The density f and cdf Φ of $X \sim \mathcal{N}(0, 1)$ being *standard normal distributed*;



`dnorm(x, 0, 1)`

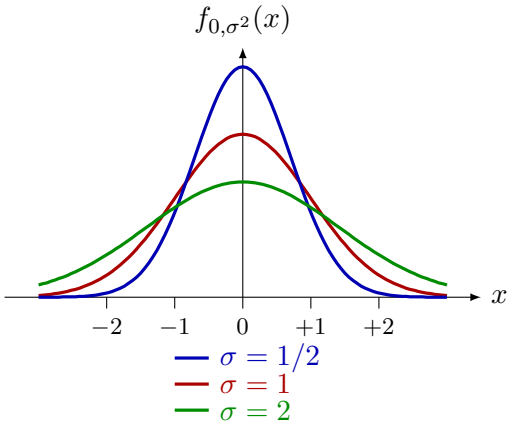


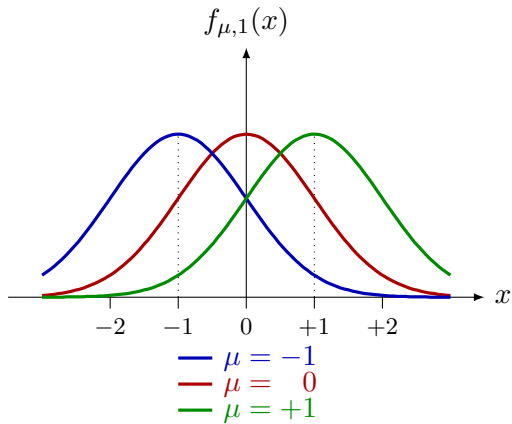
`pnorm(x, 0, 1)`

[4] Use `qnorm` for the quantile function and `rnorm` to sample from the Normal.

Normal distribution – effect of parameters μ, σ^2

Effect of mean μ and standard deviation σ as parameters for $\mathcal{N}(\mu, \sigma^2)$





Normal distribution – standard cdf

The *standard Normal* has parameters $\mu = 0$ and $\sigma^2 = 1$ with density

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

which leads to the cdf Φ of the standard normal

$$\Phi(x) = P(X \leq x) = \int_{-\infty}^{x} f(z)\,\mathrm{d}z = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-z^2/2}\,\mathrm{d}z$$

which does not have a closed form solution.

To evaluate Φ, we can use R's `pnorm` function.

Normal distribution – Properties

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ with cdf F_X and $a, b \in \mathbb{R}$, then

AFFINE TRANSFORM: $Y = aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$

STANDARDIZATION: Affine transformation to *standard location and scale*

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

This works for any random variable X for which the first two moments exist. That is $Z = \frac{X - \mathbb{E}X}{\sqrt{\mathbb{V}\mathrm{ar}(X)}}$ has mean $\mathbb{E}[Z] = 0$ and variance $\mathbb{V}\mathrm{ar}(Z) = 1$.

QUANTILES: The *p-quantile* x_p of X is related to the *standard normal p-quantile* z_p

$$x_p = \mu + \sigma z_p$$

where $z_p = \Phi^{-1}(p)$ for $p \in (0, 1)$ and $x_p = F_X^{-1}(p)$.

SYMMETRIES: For the *standard normal* density, cdf and p-quantiles holds

$$f_Z(z) = f_Z(-z), \qquad \Phi(-z) = 1 - \Phi(z), \qquad z_p = -z_{1-p}$$

χ^2-distribution

The *χ^2_n-distribution with n degrees of freedom*[5] is the *sum of n independent squared standard normal variables*. It is very *Important in Statistics*.

χ^2-Distribution

Let $X_i \overset{\text{ind}}{\sim} \mathcal{N}(0,1)$ for $i = 1, \ldots, n$, then

$$Y = \sum_{i=1}^{n} X_i^2 \sim \chi^2_k.$$

The density f of a χ^2_k random variable is

$$f(x) = \frac{x^{(n/2)-1} e^{-x/2}}{2^{n/2}\Gamma(n/2)}$$

for $x > 0$ and zero else.

[5] The pronounciaton of χ^2 is "chi-squared"

Student's t-distribution

Student's *t-distribution with n degrees of freedom* occures as a scaled *ratio of a standard normal* and the *root of a χ^2_n-distribution*. *Important in Statistics*.

t-Distribution

Let $U \sim \mathcal{N}(0,1)$ and $V \sim \chi^2_n$ independent, then

$$X = \frac{U}{\sqrt{V/n}} \sim t_n.$$

The density f of a t_n random variable is

$$f(x) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{\pi n}\Gamma\left(\frac{n}{2}\right)}\left(1+\frac{x^2}{n}\right)^{-(n+1)/2}$$

for $x \in \mathbb{R}$.

More Continuous distribution

To name a few other common distributions

Hypergeometric

Wishart	negative Binomial
Truncated Normal	log-Normal
Dirichlet	Cauchy
Gamma	F-distribution
Beta	Inverse χ^2

and many many more ...